YASHAS SRINIVASAN KADAMBI

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EDUCATION

PES UNIVERSITY **Bengaluru, India**
Bachelor of Technology with Distinction in Computer Science and Engineering **Aug 2019 - May 2023**

- GPA: 9.33

CITY PU COLLEGE **Bengaluru, India**
II PU in PCMC **Jul 2018 - Mar 2019**

- Grade: 93 %

PUBLICATIONS

Ramesha, V. et al. (2024). "Towards Faster and Efficient Lightweight ISR Using Transformers and Fourier Convolutions" in Artificial Intelligence and Applications, Jan 2024. doi: 10.47852/bonviewAIA42021930.

- Designed and implemented a **hybrid Transformer and FFC (Fast Fourier Convolutions)** model for Image Super Resolution that has **34% fewer parameters** while being **60% faster.**
- Created the Dual Frequency Spatial block to **selectively glean features from both the spatial and frequency domains**, resulting in significantly higher visual quality of images containing complex patterns.

Kadambi, Yashas et al. (2021). **"Monitoring and Alert Systems for Underwater Data Centers using Arduino".** In: 2021 IEEE Xplore (CSITSS), pp. 1–6. doi: 10.1109/CSITSS54238.2021.9683449.

- Proposed and designed a resilient Arduino-based IOT device for monitoring underwater assets; built with **redundancy and high availability,** considering the expensive and logistically challenging maintenance.

PROJECTS

- **Performance Analysis of the GHOST Scheduler**
  - **Rebuilt the Linux kernel** to make use of Google's ghOSt scheduler, allowing the user-space processes access to the kernel's scheduling policies.
  - Evaluated the performance of the **Completely Fair Scheduler, FIFO, and shinjuku schedulers** with and without the ghOSt framework on RocksDB and on a backend server in various scenarios like burst/prolonged load, 16/32 threads, and 16 GB/32 GB RAM.
- **Distributed Machine Learning for Image Classification (CIFAR-10)**
  - Designed and implemented an **Apache Spark application to stream images** between servers and trained an ML model to classify these images.
  - Analyzed the **impact of various streaming batch sizes on image classification precision and accuracy** and optimized the streaming workflow for the best performance.

PROFESSIONAL EXPERIENCE

CISCO SYSTEMS **Bengaluru, India**
Optical Software Development Engineer **Jan 2025 - Present**

Developed software for the line-cards used on NCS 1014 series of routers

- Designed and implemented a comprehensive Unit testing framework, **cutting UT time for all new features by over 90%** and enabling higher velocity for feature delivery.
- Delivered QPSK modulation in the Line card for increased performance and reach for long-distance applications.
- Developed the end-to-end software in the Data-plane of the new Aquila line card, including **integrating the CDR (Clock and Data Recovery)** hardware with the line card and integrating a secure boot process.
- Created a tool to extract Performance Monitoring Data from the logs collected on routers deployed in the customer network for faster debugging.
- Identified and created an internal tool automating routine tasks like building and deployment of the new binaries and **streaming the device logs in near real time to the local machines**, significantly increasing developer productivity.

Software Development Engineer **Aug 2023 - Jan 2025**

Worked as a Backend Engineer for the Px Cloud, SNTC and supporting applications

- Developed Terraform templates for AWS EC2, ECS, Lambdas, RDS, DynamoDB, SQS, and SNS and IAM Services. I **investigated performance bottlenecks** and parallelised the deployment plan to cut deployment time by 50%
- Owned and enhanced the common modules of Terraform, **enabling consistency and efficiency in the migration** of all the services in PX Cloud to Infrastructure-as-code
- Configured **event triggers for DynamoDB, SQS, and SNS across multiple AWS services** to automate deployment workflows of interdependent components.
- **Led the migration from Ping authentication to Okta** for a legacy product, ensuring no production impacts across seven different products by conducting knowledge transfer sessions and tracing consumer usage.
- **Found and fixed the root cause of 4 Production outages** in collaboration with the operations and infrastructure teams; further led the investigation into why these outages occurred and created a plan to eliminate further outages.
- Identified and resolved Performance issues spanning across Postgres, Mongo, and Cassandra Databases, decreasing **page load time by 40% for an application having 2 Billion USD annual revenue**.
- **Created a QA chatbot using OpenAI APIs and Retrieval-Augmented Generation (RAG)** to actively capture and store messages from topic-specific group chats (e.g., CI/CD group), leveraging these stored messages as dynamic context to generate up-to-date and accurate responses with the LLM.
- **Mentored two new joinees to the team by providing KT sessions,** outlining the code-quality expectations, and helping them ramp up to the new project.

Technical Intern **Jan 2023 - Jun 2023**

- Overhauled the PXCloud Client SDK, establishing a CI pipeline for automatic generation and publication of the Python SDK, **eliminating 4-hour manual updates**, and ensuring the latest SDK versions are generated.
- Implemented an alternative to Docker Desktop by automating installation on individual machines, **saving $1,500 in the annual budget** for the team.

Schneider Electric **Bengaluru, India**

Summer Intern **May 2022 - Jul 2022**

- Developed a software tool independently through requirement engineering, architecture design, and development, **reducing test case discussions by over 90%**, from two days to two hours, significantly improving velocity.

HIGHLIGHTS

- Technical Skills: Java, C, Python, AWS, JavaScript, Terraform
- **AWS Certified Developer Associate**
- Awarded the Prof MRD scholarship for being in top 20% of the CSE department
- Member of the Official Cisco Capture The Flag (CTF) team.
- Winner of the Cybersecurity CTF (with a participation of over 2000 members)
- Developed a multilingual voice assistant using Retrieval-Augmented Generation, optimized to run effectively without a GPU (~20 seconds per query), enabling access for farmers to governmental subsidy and relief resources.

TEACHING ASSISTANT

Image Processing and Computer Vision **Dec 2022 - May 2023**

- Developed a **series of labs with answers steganographically embedded** in images to encourage students to apply classical image processing techniques practically and creatively.
- Wrote a custom Excel macro that partially **automated a laborious manual grading** process, improving efficiency.

Data Analytics **Aug 2022 - Dec 2022**

- **Designed TV-show themed labs and assignments** for improving engagement and improved access to help by hosting office hours and class forums.

Deep Learning **Jul 2022 - Dec 2022**

- Created teaching materials, assignments, and labs focusing on Autoencoders and Generative Adversarial Networks